Robotics, Artificial Intelligence and Embedded Systems

Cognitive Systems: Neural Networks and Applied AI

Part II: Neural Computation

Chapter 5
Deep Neural Networks and the Brain

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Florian Walter, M.Sc.

Topics in this Chapter

How is deep learning related to the brain?

- Why are feature space representations important?
- How can deep neural networks learn features automatically?
- Which properties do deep convolutional neural networks share with the human visual system?
- What are the limitations of current deep neural network architectures?

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Basic Machine Learning – An Example

Classifying Iris Species

Iris is a genus of about 300 species of plants that produce a very high variety of different blossoms. The Iris flower dataset from 1936 contains data from three different species and has become one of the standard tests for machine learning algorithms.

Iris setosa

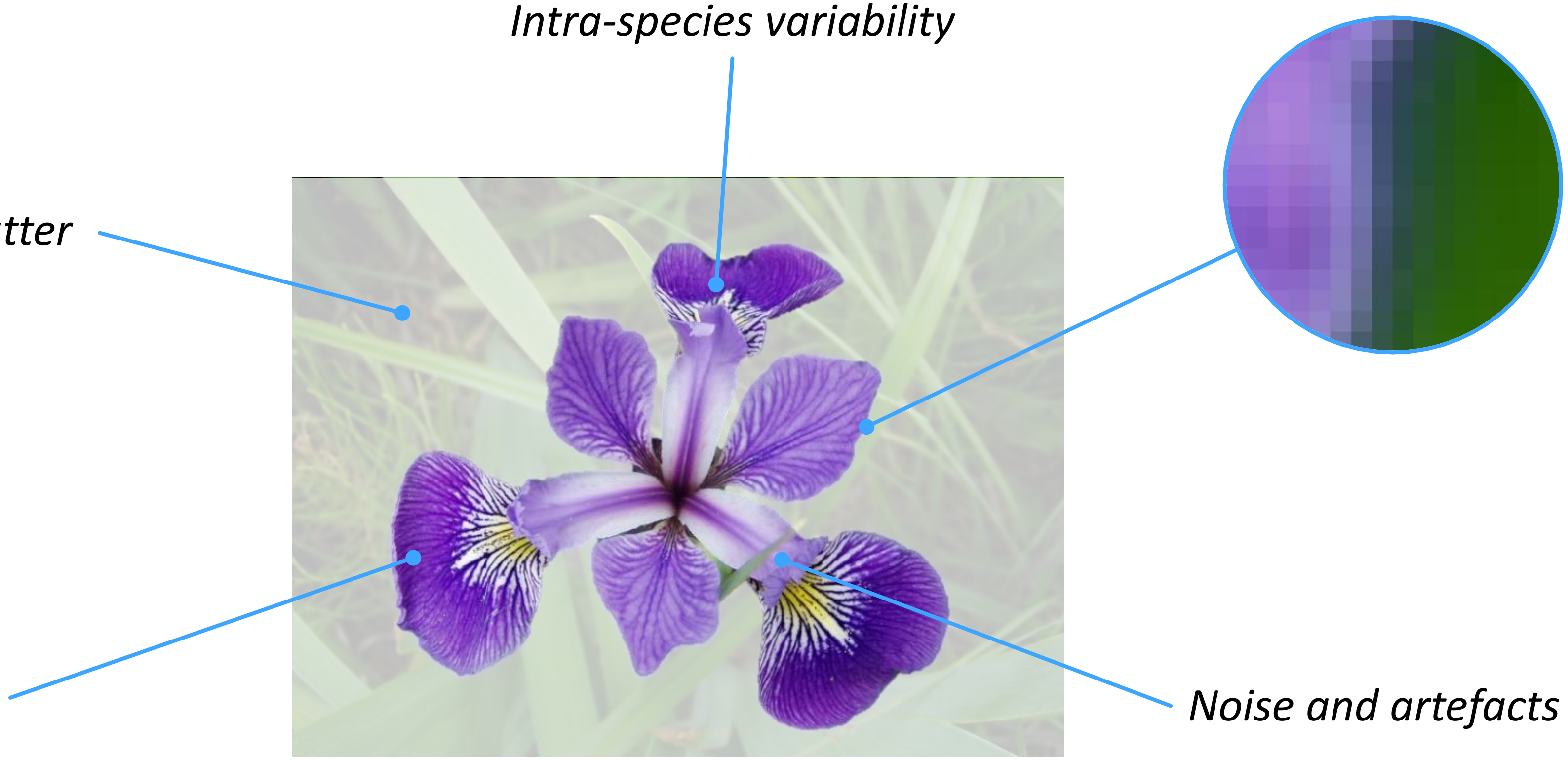
Iris versicolor

Iris virginica

Image Source: Wikimedia Commons

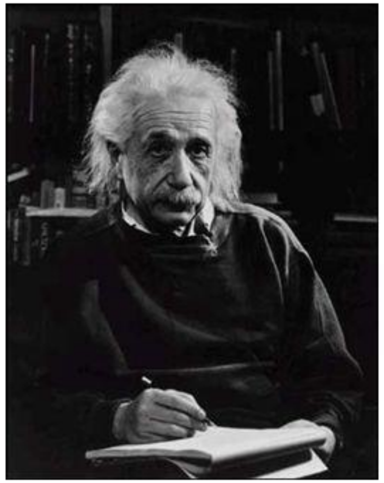
Working with Real-World Data

Processing images is challenging



What does Data Look Like for the Computer?

Computer Vision



What we see

0	3	2	5	4	7	6	9	8
3	0	1	2	3	4	5	6	7
2	1	0	3	2	5	4	7	6
5	2	3	0	1	2	3	4	5
4	3	2	1	0	3	2	5	4
7	4	5	2	3	0	1	2	3
6	5	4	3	2	1	0	3	2
9	6	7	4	5	2	3	0	1
8	7	6	5	4	3	2	1	0

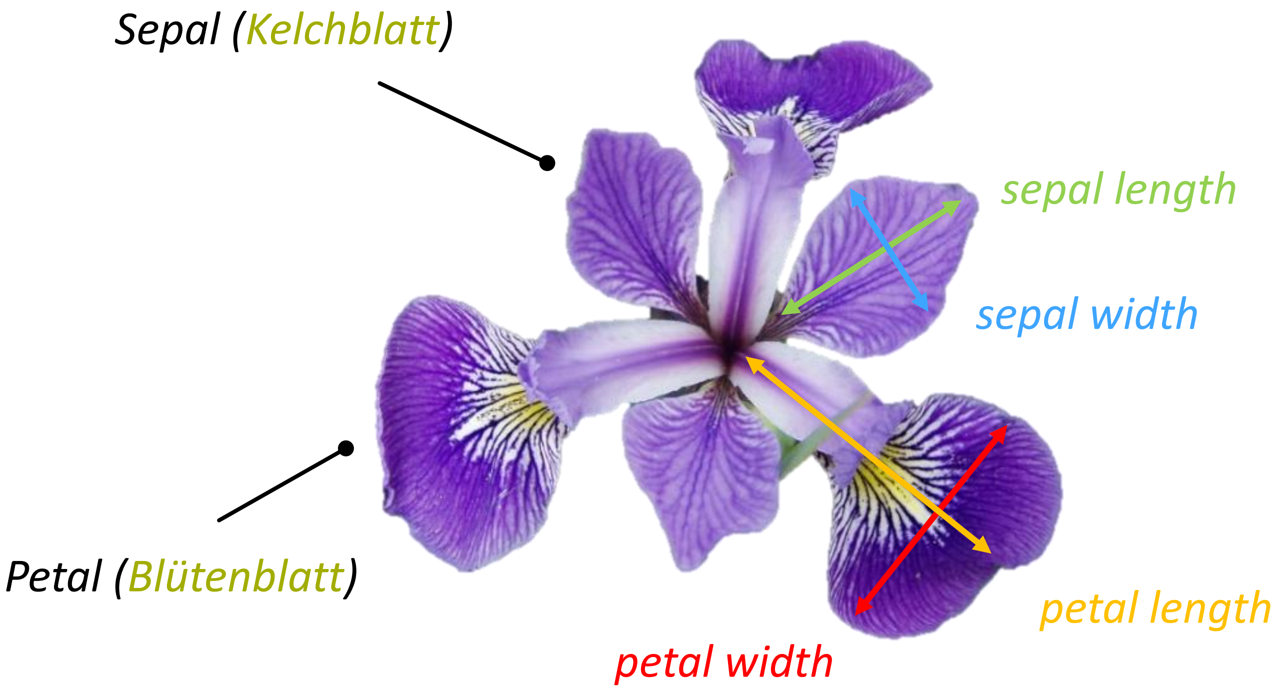
What a computer sees

https://medium.com/@eternalzer0dayx/demystifying-convolutional-neural-networks-ca17bdc75559

Feature Representation

Features (Merkmale) abstract from the raw-data and represent semantic information in the data

- In the iris dataset, the blossom of every individual of a species is represented by four measures (sepal length, sepal width, petal length, petal width)
- These features encode the information processed by the machine learning algorithm
- Designing features is one of the most challenging tasks in machine learning



What Does the Iris Dataset Look Like?

- The features of all samples in the dataset (i.e. flowers that have been collected and measured) are stored in a table
- Every row in the table corresponds to one individual
- The Iris flower dataset contains a total of 150 samples with 50 samples for each of the three considered species
- Modern real-world datasets are multiple orders of magnitude larger and usually contain more features!

#	Sepal Length	Sepal Width	Petal Length	Petal Width	Class
10	5.4	3.7	1.5	0.2	Iris-setosa
32	5.2	4.1	1.5	0.1	Iris-setosa
42	4.4	3.2	1.3	0.2	Iris-setosa
47	4.6	3.2	1.4	0.2	Iris-setosa
51	6.4	3.2	4.5	1.5	Iris-versicolor
70	5.9	3.2	4.8	1.8	Iris-versicolor
84	5.4	3.0	4.5	1.5	Iris-versicolor
96	5.7	2.9	4.2	1.3	Iris-versicolor
99	5.7	2.8	4.1	1.3	Iris-versicolor
100	6.3	3.3	6.0	2.5	Iris-virginica
101	5.8	2.7	5.1	1.9	Iris-virginica
107	7.3	2.9	6.3	1.8	Iris-virginica
110	6.5	3.2	5.1	2.0	Iris-virginica

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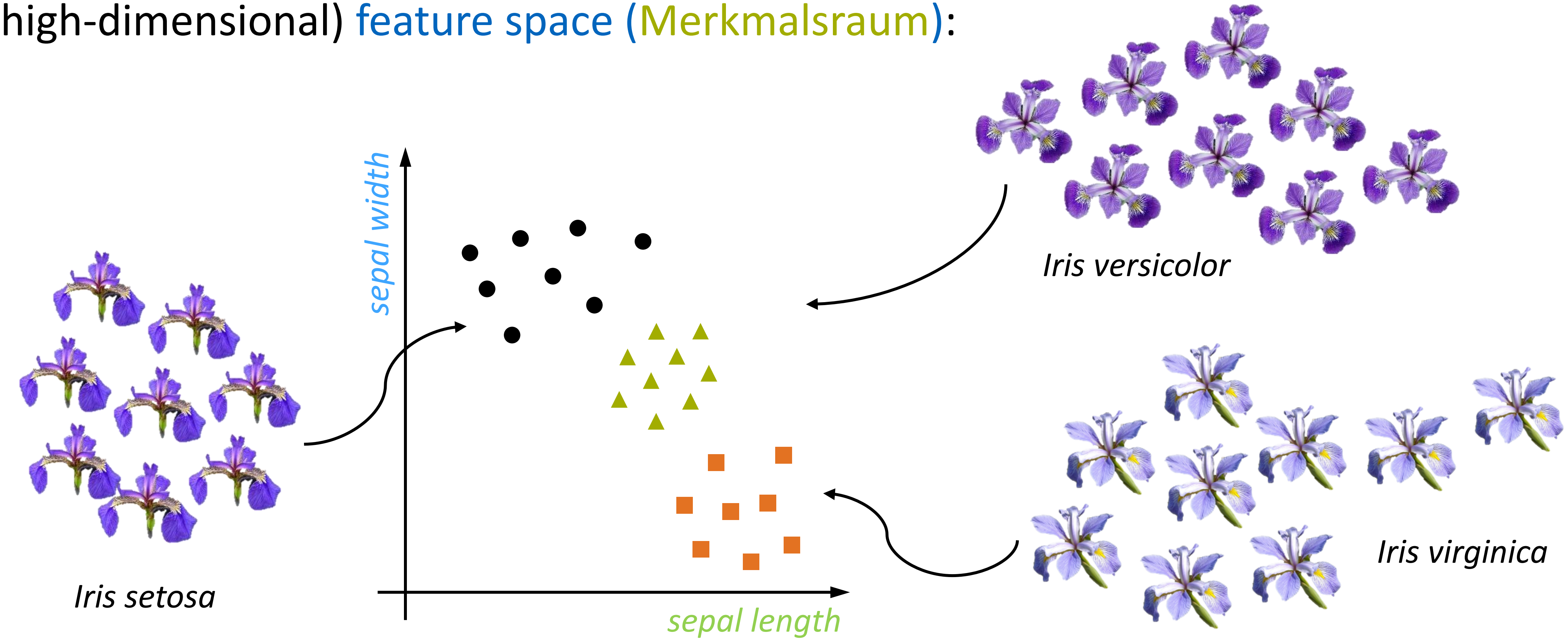
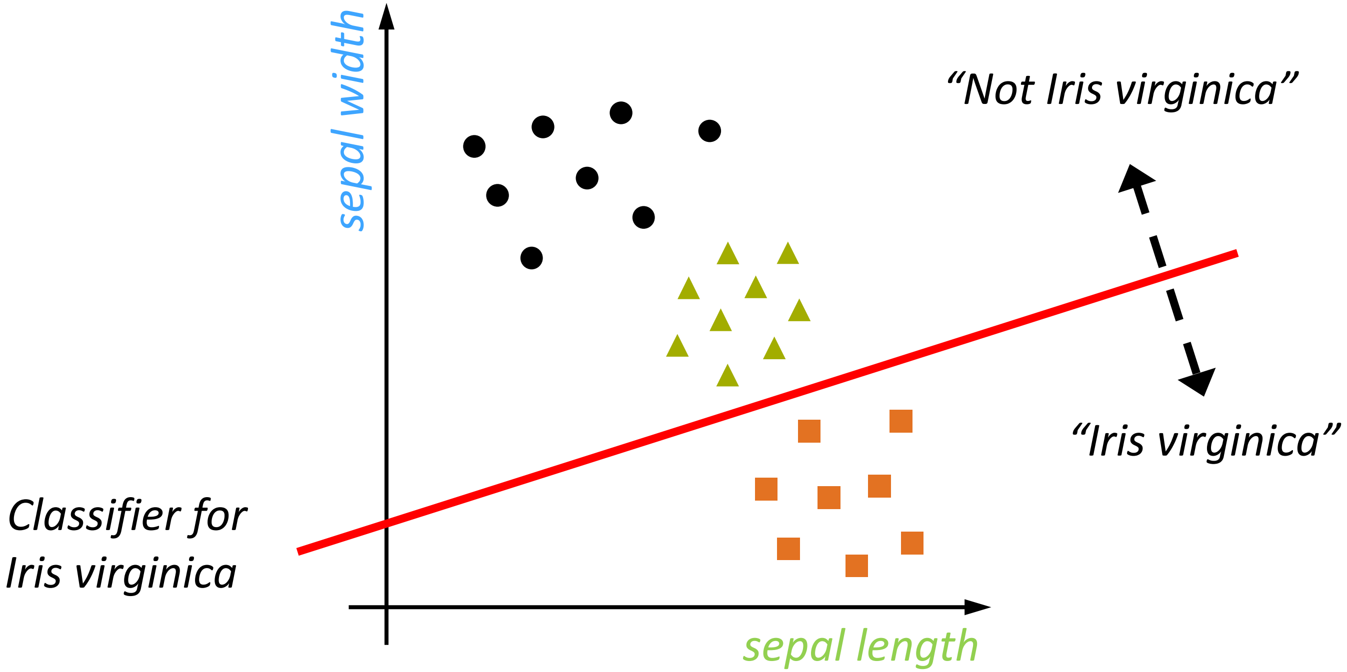
Samples in the Feature Space

Based on the features in the dataset, every sample of an Iris blossom can be represented in a (usually high-dimensional) feature space (Merkmalsraum):



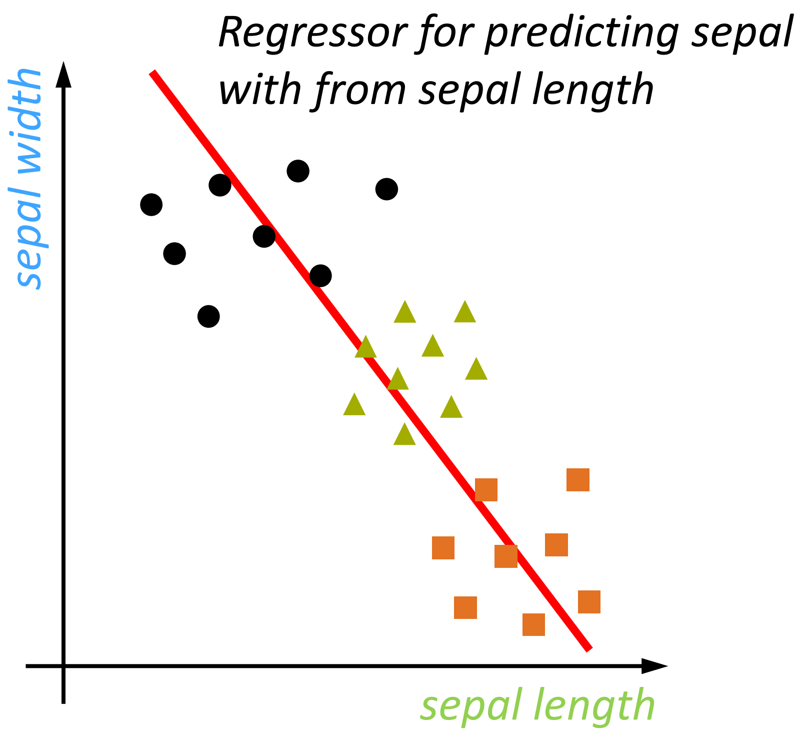
Classification

In machine learning, classification means assigning a sample to one of *n* discrete classes. If a suitable feature space representation is available, two classes can be easily separated by a line:

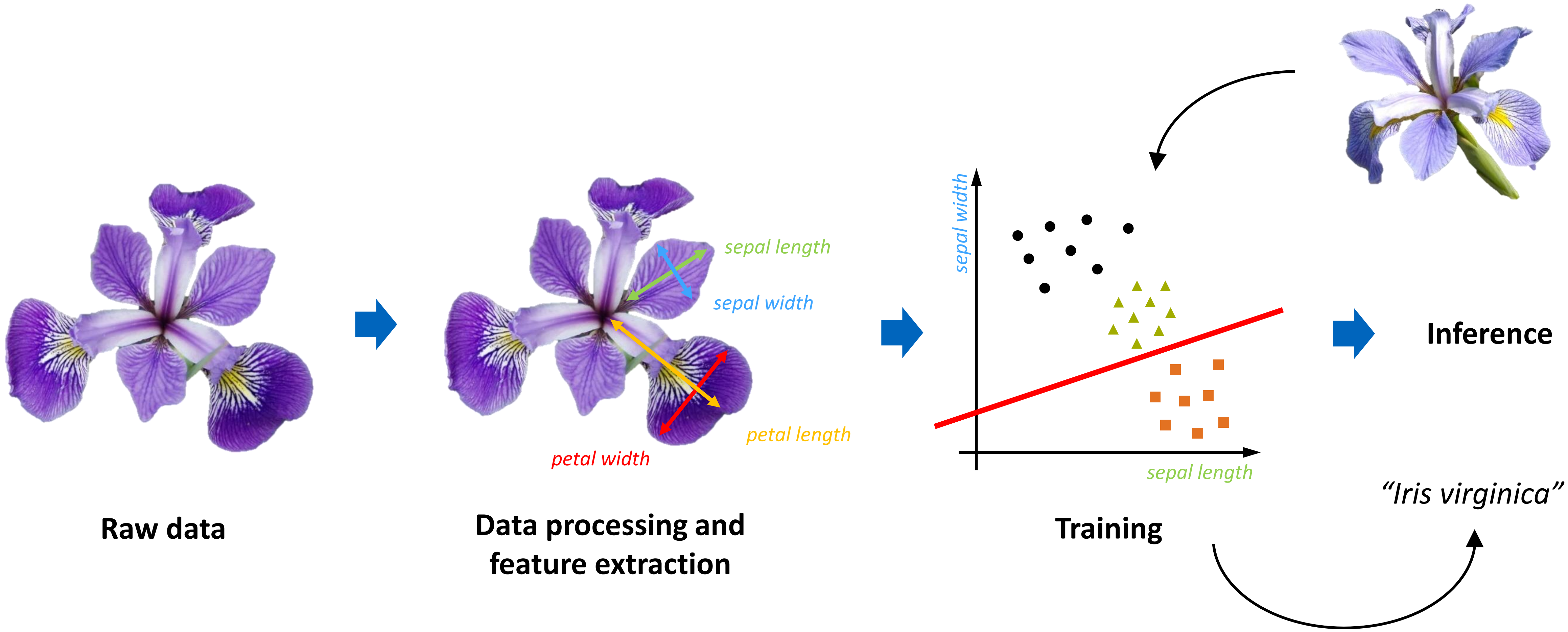


Regression

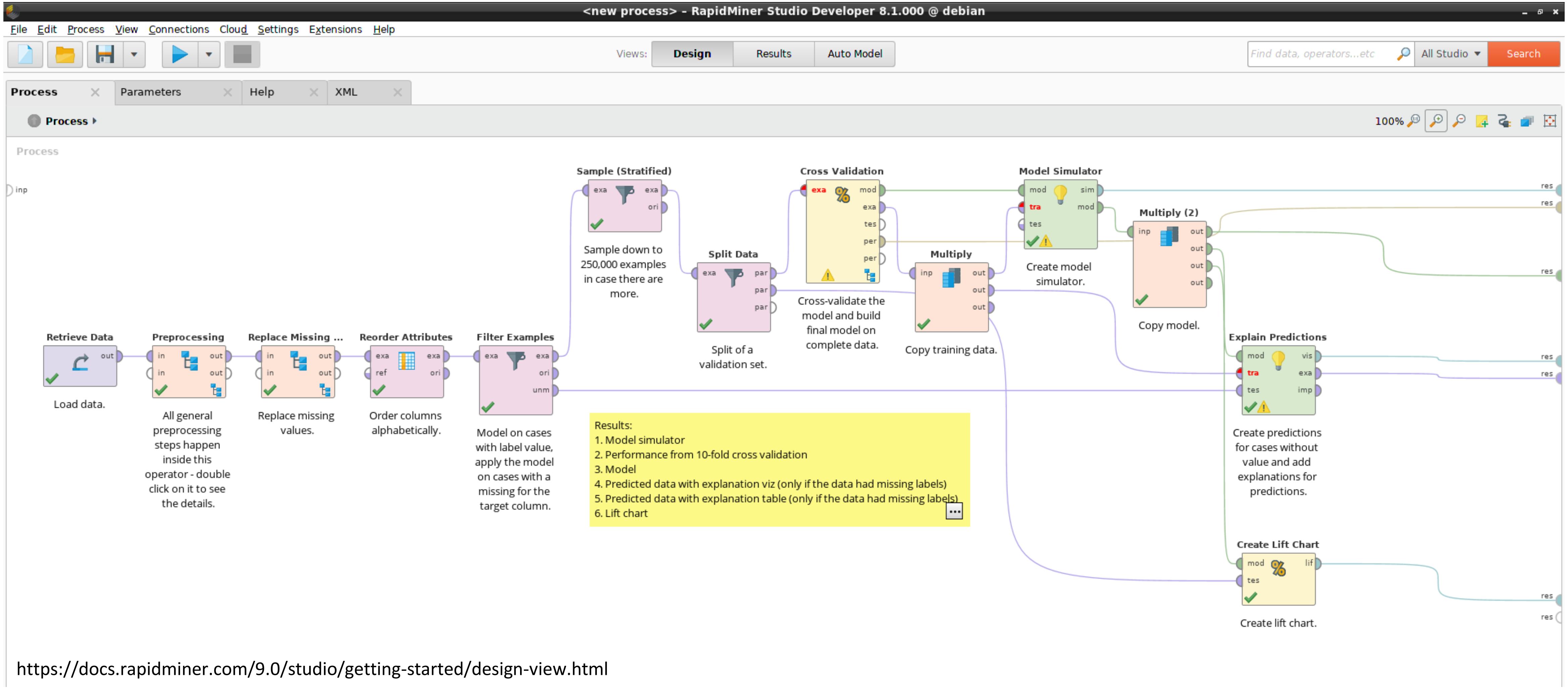
In machine learning, regression means assigning a continuous value to a sample. This can, for example, be used to predict missing points of a curve.



Summary: The Typical Workflow in Machine Learning



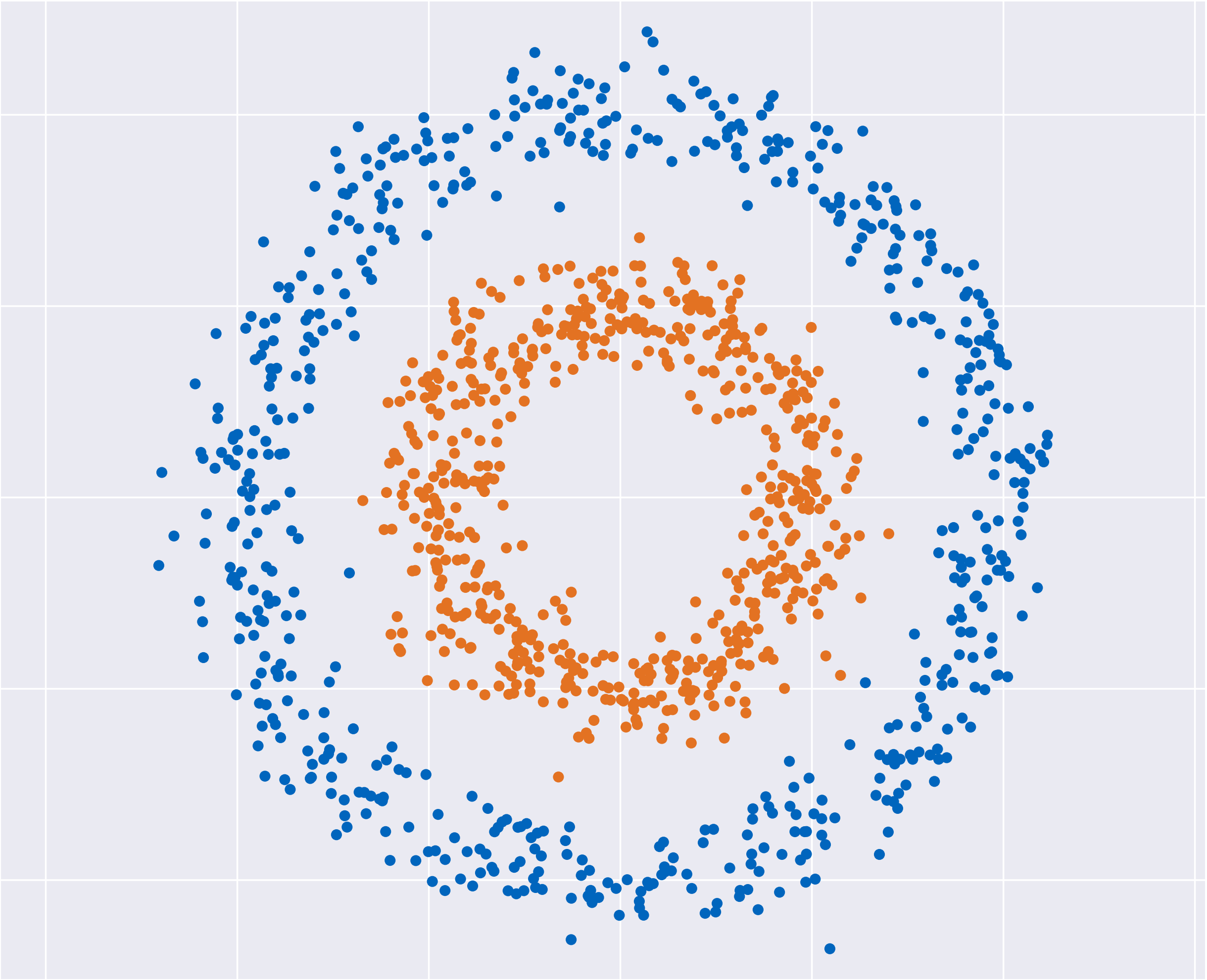
Example: Workflow in a Data Science Application



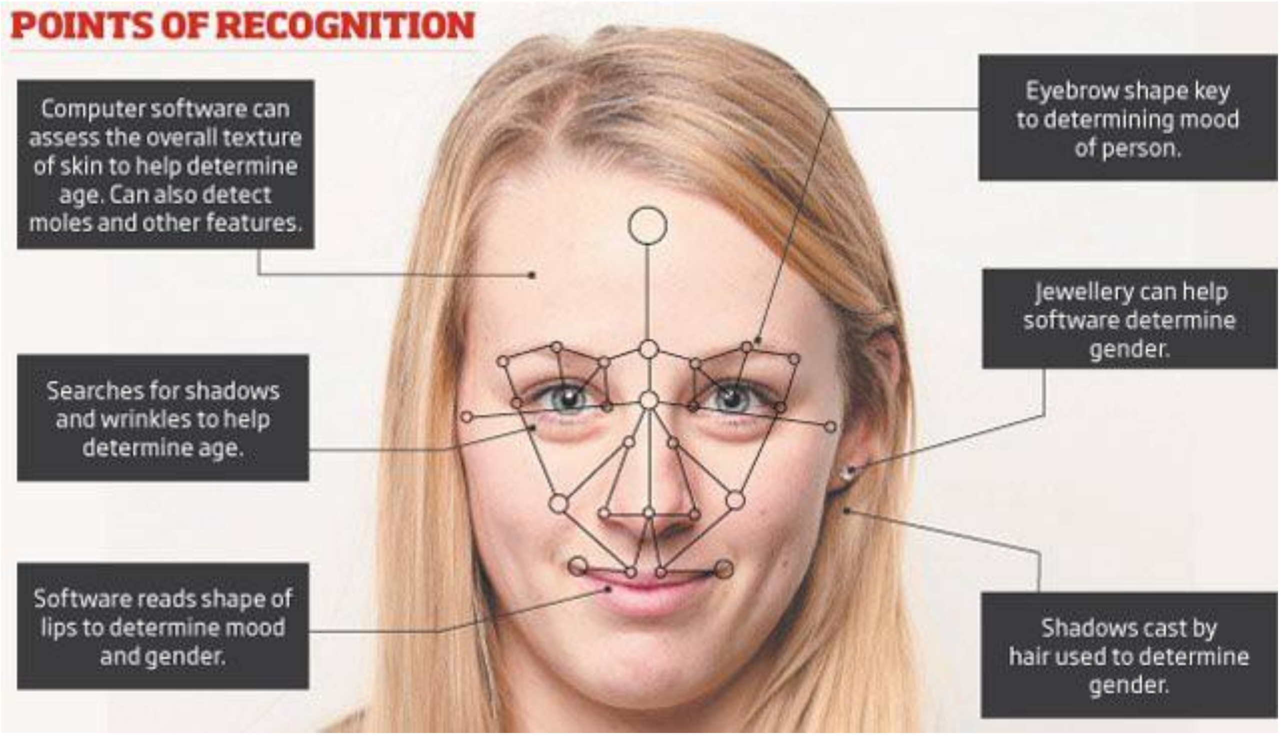
https://docs.rapidminer.com/9.0/studio/getting-started/design-view.html

The Challenge

- In general, real-world data sets are not linearly separable (such as the XOR problem)
- Applying linear classifiers is therefore not possible in the original input space
- An essential part in traditional machine learning is therefore to map the data into a feature space (Merkmalsraum) in which they are linearly separable (feature engineering)



What is the Right Feature Space for Faces?



https://www.tractiontechnology.com/blog/gaining-traction-in-ai-machine-learning-and-data

Feature Space Representations

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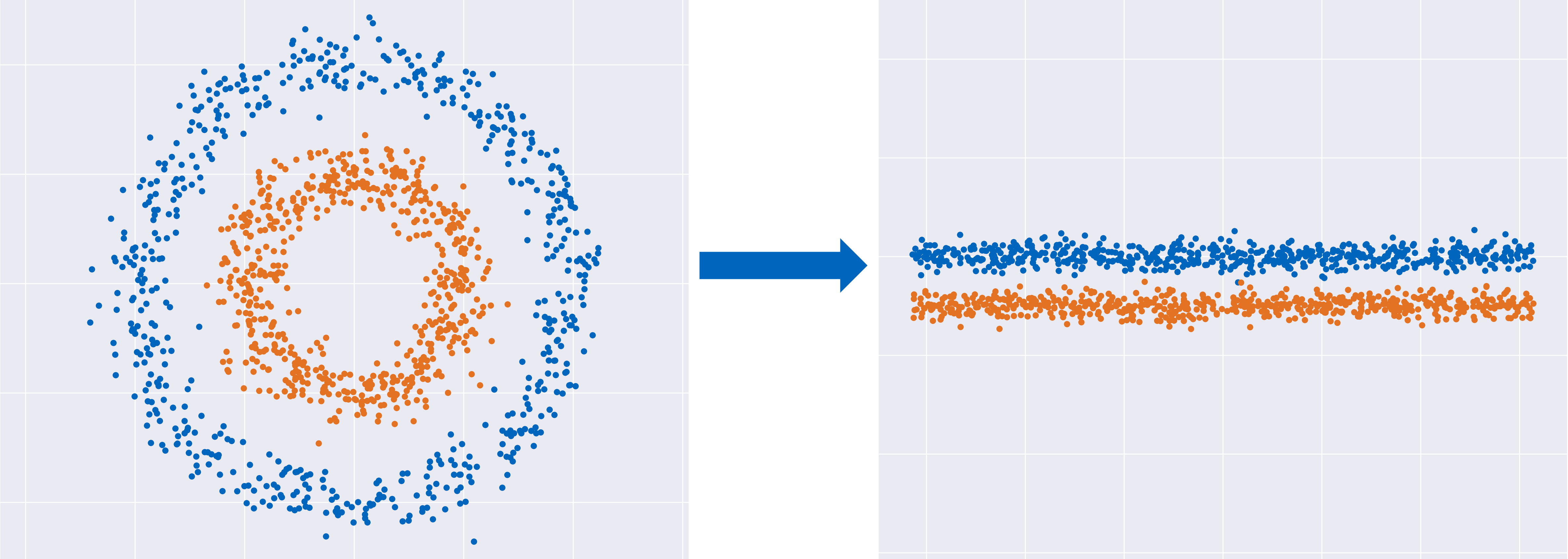
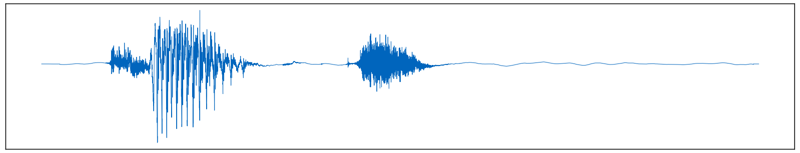
Example: Simple Coordinate Transform



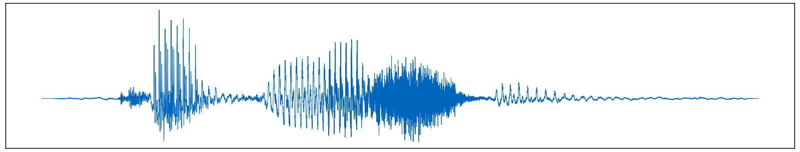
Cartesian Coordinates

Polar Coordinates

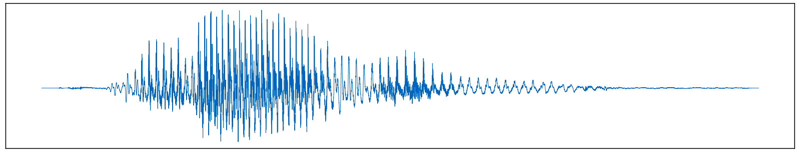
Example: Labeled Audio Files



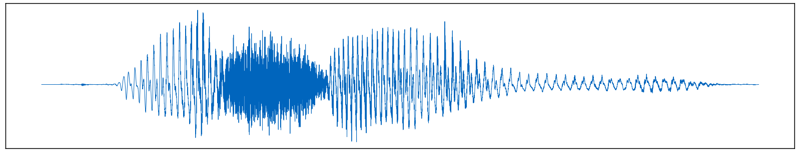
cat



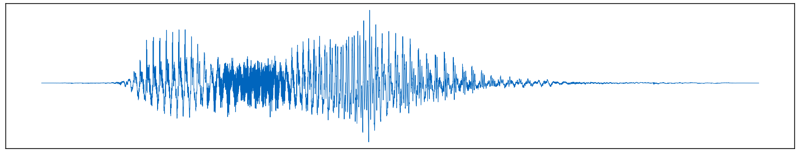
cognition



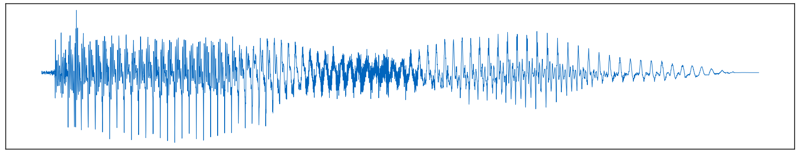
learning



machine



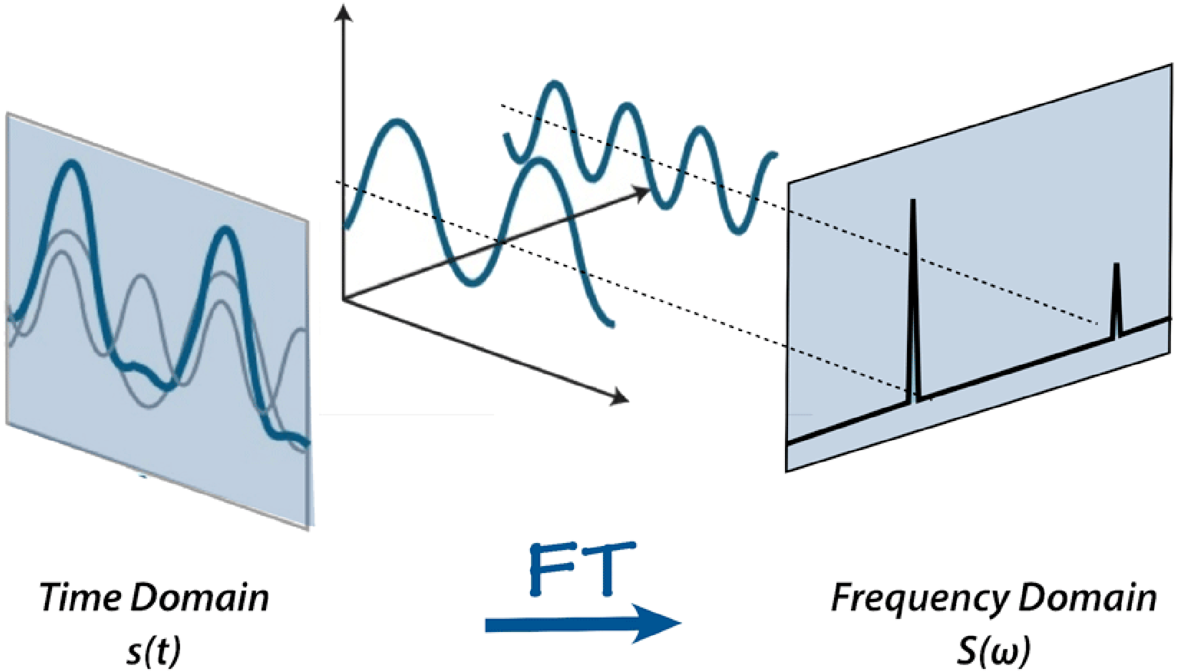
azure



azure

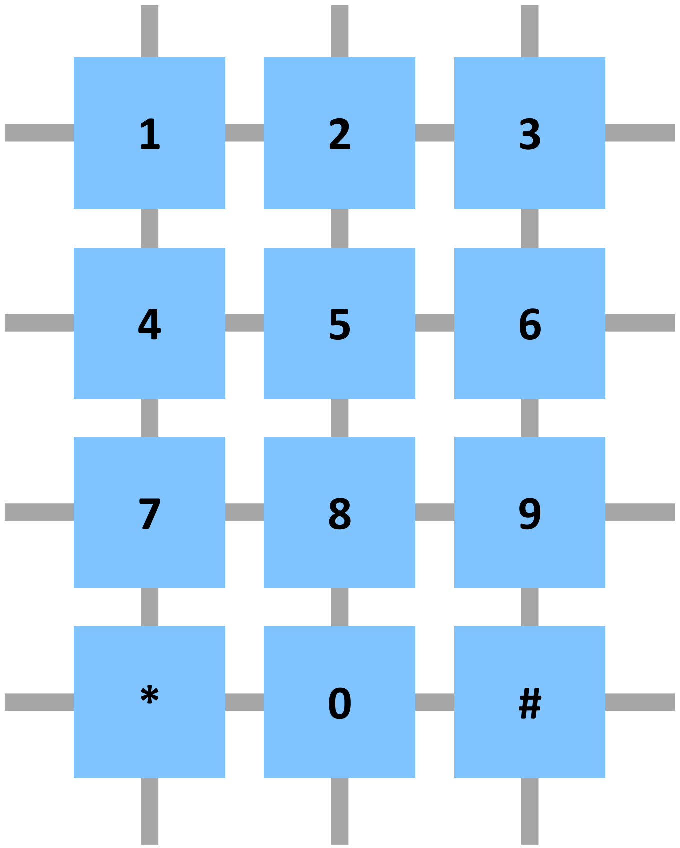
Audio from http://www.leo.org/

Fourier Transform for Speech Recognition

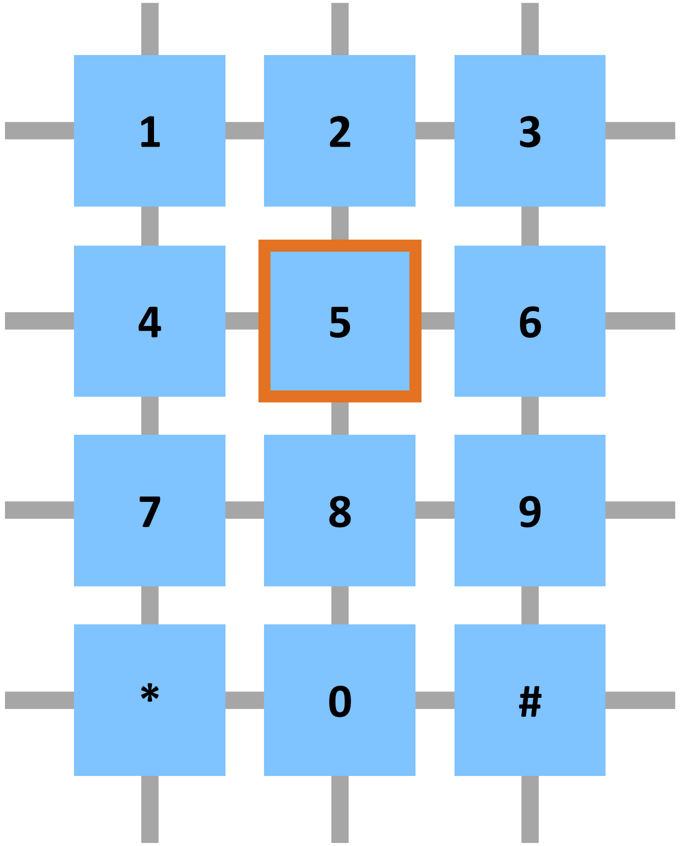


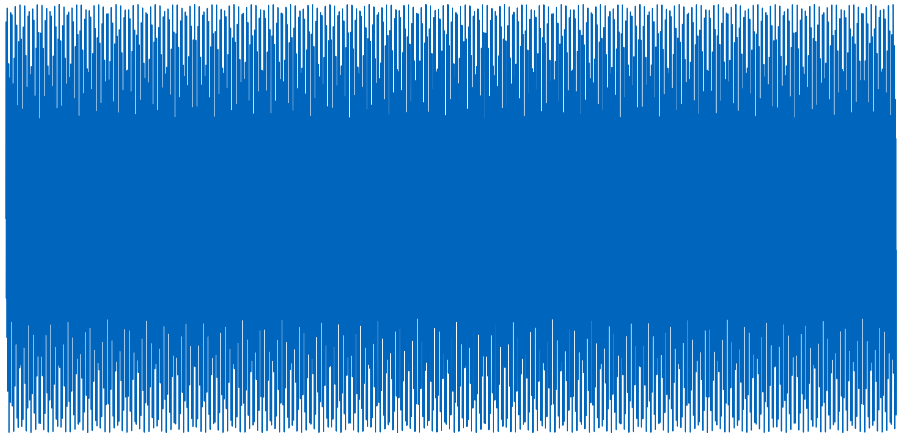
http://mriquestions.com/fourier-transform-ft.html

Example: Dual-Tone Multi-Frequency Signaling



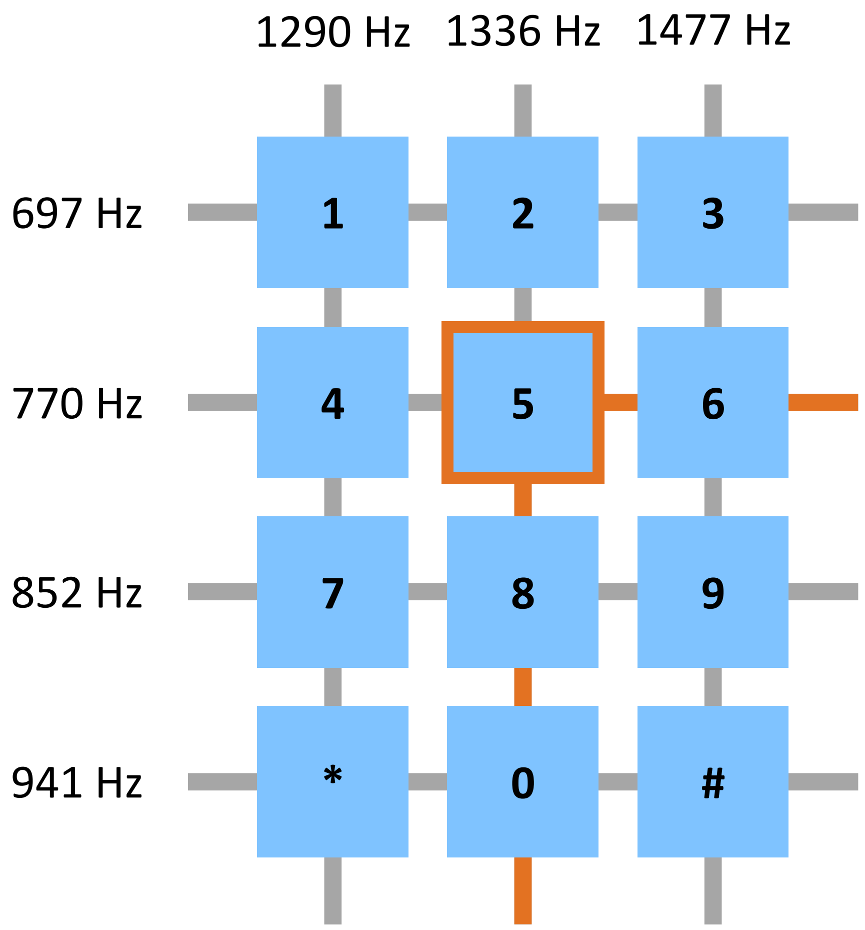
Example: Dual-Tone Multi-Frequency Signaling

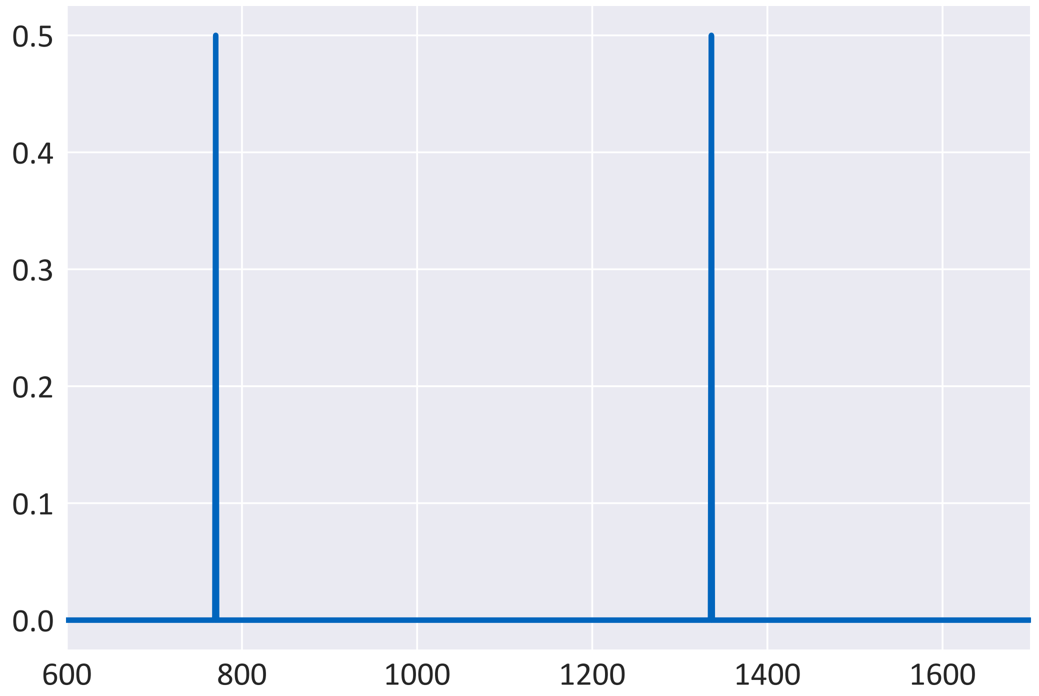




Waveform representation

Example: Dual-Tone Multi-Frequency Signaling

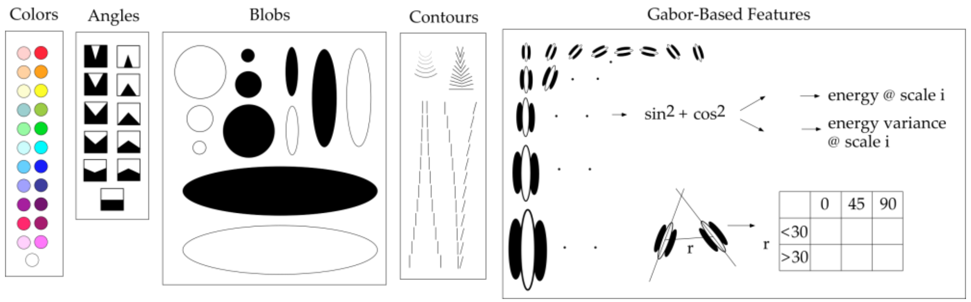




Representation in frequency space after Fourier transform

Example: SEEMORE

Figure 1: The database contains 100 objects of many different types, including rigid (soup can), nonrigid (necktie), statistical (bunch of grapes), and photographs of complex indoor and outdoor scenes.



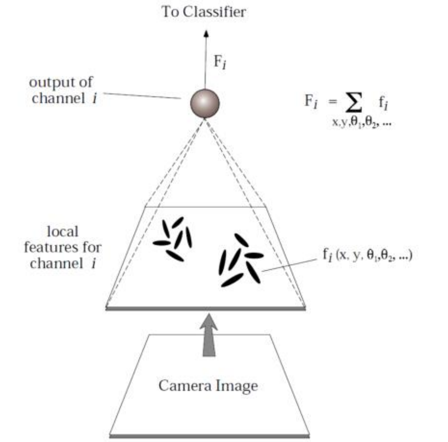


Figure 4: The ith feature channel is based on an elemental nonlinear filter $f_i(x, y, \theta_1, \dots)$, which is subsampled across the image, at a range of image-plane orientations, and over any of the filters' additional internal degrees of freedom. The output of the ith channel F_i is the sum over all elemental filter outputs within that channel.

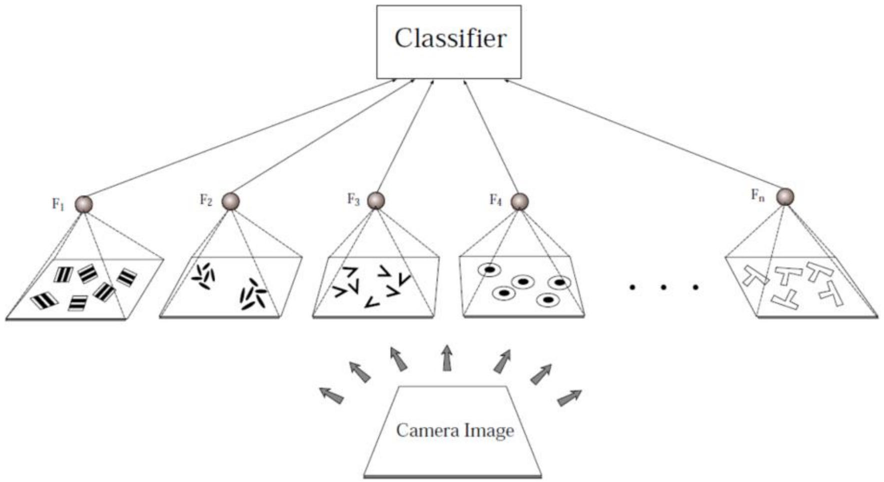


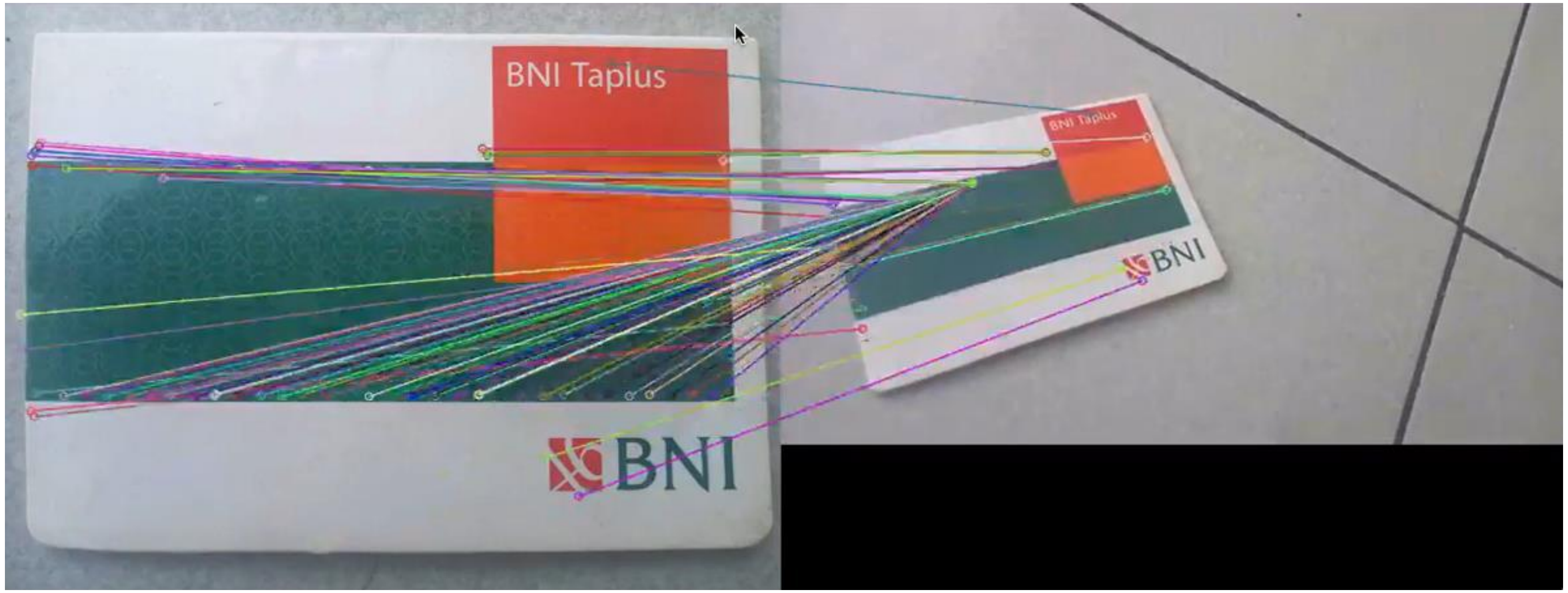
Figure 5: SEEMORE's architecture consists of a set of channels that provides input to a nearest-neighbor classifier. Since each channel is by itself insensitive to translation and rotation in the image plane and is slowly varying as an object rotates in depth, the output vector **F** for the complete set of channels is similarly invariant to these transformations. However, **F** remains highly sensitive to changes in object quality, and hence object identity.

Bartlett W. Mel, SEEMORE: Combining Color, Shape, and Texture Histogramming in a Neurally Inspired Approach to Visual Object Recognition, Neural Computation 9(4):777-804 · June 1997

Example: SIFT

SIFT (David Lowe, 1999): Automatic identification of image key points and computation of image features that are invariant to with respect to scaling, image translation and rotation

Matching of image key points detected by SIFT



Video adapted from https://youtu.be/uA7Yxu502uw

SIFT and the Brain

- The SIFT features are designed to be invariant to scale, translation and partially invariant to illumination changes and projections
- Specific neurons in the inferior temporal cortex in primates share similarities with the SIFT features
- Their activity is invariant to changes in scale, location and illumination

Talk by David G. Lowe: https://youtu.be/x35bHNvkK1o

SIFT with OpenCV: https://youtu.be/USl5BHFq2H4

Object Recognition from Local Scale-Invariant Features

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Proc. of the International Conference on Computer Vision, Corfu (Sept. 1999)

Abstract

An object recognition system has been developed that uses a new class of local image features. The features are invariant to image scaling, translation, and rotation, and partially invariant to illumination changes and affine or 3D projection. These features share similar properties with neurons in inferior temporal cortex that are used for object recognition in primate vision. Features are efficiently detected through a staged filtering approach that identifies stable points in scale space. Image keys are created that allow for local geometric deformations by representing blurred image gradients in multiple orientation planes and at multiple scales. The keys are used as input to a nearest-neighbor indexing method that identifies candidate object matches. Final verification of each match is achieved by finding a low-residual least-squares solution for the unknown model parameters. Experimental results show that robust object recognition can be achieved in cluttered partially-occluded images with a computation time of under 2 seconds.

1. Introduction

Object recognition in cluttered real-world scenes requires local image features that are unaffected by nearby clutter or partial occlusion. The features must be at least partially invariant to translation, scaling, and rotation, and partially invariant to illumination changes and affine or 3D projection. Previous approaches to local feature generation lacked invariance to scale and were more sensitive to projective distortion and illumination change. The SIFT features share a number of properties in common with the responses of neurons in inferior temporal (IT) cortex in primate vision. This paper also describes improved approaches to indexing and model verification.

The scale-invariant features are efficiently identified by using a staged filtering approach. The first stage identifies key locations in scale space by looking for locations that are maxima or minima of a difference-of-Gaussian function. Each point is used to generate a feature vector that describes the local image region sampled relative to its scale-space coordinate frame. The features achieve partial invariance to local variations, such as affine or 3D projections, by blurring image gradient locations. This approach is based on a model of the behavior of complex cells in the cerebral cortex of mammalian vision. The resulting feature vectors are called SIFT keys. In the current implementation, each image generates on the order of 1000 SIFT keys, a process that requires less than 1 second of computation time.

The SIFT keys derived from an image are used in a nearest-neighbour approach to indexing to identify candidate object models. Collections of keys that agree on a po-

Summary: Feature Engineering

- Feature spaces are highly domain dependent
- The performance of a learning algorithm is highly dependent on the quality of the feature space representation
- Manual feature engineering can become extremely complex for real-world datasets